

LimitIQ

Dynamic Credit Line Management & Exposure Optimization

Executive decision brief

LimitIQ evaluates current limit, +10%, +20% and +30% candidates, then selects the highest eligible simulated risk-adjusted contribution under loss, exposure, payment-history, overextension and profitability controls. Deteriorating accounts are frozen or referred rather than automatically decreased.

Evidence class	Result
Observed source	30,000 Taiwan credit-card accounts; April–September 2005 behavior
Model estimate	Untouched-test ROC-AUC 0.781; Brier 0.133
Simulation	288 increases; INR 2,979,958 contribution

Portfolio simulation

Accounts	1,200
Current / proposed limits	INR 478,946,605 / INR 513,032,327
Current / proposed EAD	INR 401,899,201 / INR 427,463,492
Current / proposed expected loss	INR 48,711,764 / INR 50,339,593
Simulated incremental contribution	INR 2,979,958
Simulated risk-adjusted return	11.66%
Early-warning / manual review	797

Model governance

Champion: Histogram gradient boosting. The model type and threshold (0.174) were selected using validation data before one untouched-test evaluation. Sex, education, marital status, age, customer ID and target are excluded from decisioning.

Boundaries and committee actions

All line response, drawdown and financial effects are deterministic simulations. PD is not causal. Source TWD monetary values are converted to INR at the documented fixed rate for presentation; this is not Indian borrower evidence. A production build needs current local-law review, verified ability-to-pay inputs, outcome monitoring, model validation, controlled overrides, change approval and rollback. The displayed management ECL is not an IFRS 9 allowance or regulatory-capital calculation.

LimitIQ is an educational portfolio demonstration using public and synthetic data. It is not a production credit-decision system and must not be used to make real lending decisions.